UNIVERSITY OF MORATUWA
Faculty of Engineering
Department of Electronic and Telecommunication Engineering
B.Sc. Engineering
Semester 7 (Intake 2021) Examination

EN3330 INTRODUCTION TO ENGINEERING OPTIMIZATION

Time allowed: **Two (2)** hours November 2025

INSTRUCTIONS TO CANDIDATES:

- This paper contains 4 questions on 5 pages.
- This examination accounts for **30**% of the module assessment. The total maximum mark attainable is **100**. The marks assigned for each question & sections there of are indicated in brackets.
- This is a closed-book examination.
- Electronic/communication devices are not permitted. Only equipment allowed is a calculator approved and labeled by the Faculty of Engineering.
- Derivations are not required if they are not explicitly requested for in the question.
- Assume reasonable values for any data not given in or with the examination paper. Clearly state such assumptions.
- If you have any doubt as to the interpretation of the wording of a question, make your own decision, and clearly state it.
- All questions carry equal marks.
- Answer **ALL** questions.
- Keep your answers **brief** and to the point.

ADDITIONAL MATERIAL:

- No additional materials are provided.

Continued ...

Question 1.

(a) Seven critical points of a function $f(x)$, which is defined on the closed interval $x \in \mathbb{R} : -5 \leq x \leq 15$ is shown in Figure Q1.1. State whether each critical point corresponds to a minimum (local/global), a maximum (local/global), or neither. **[5 marks]**

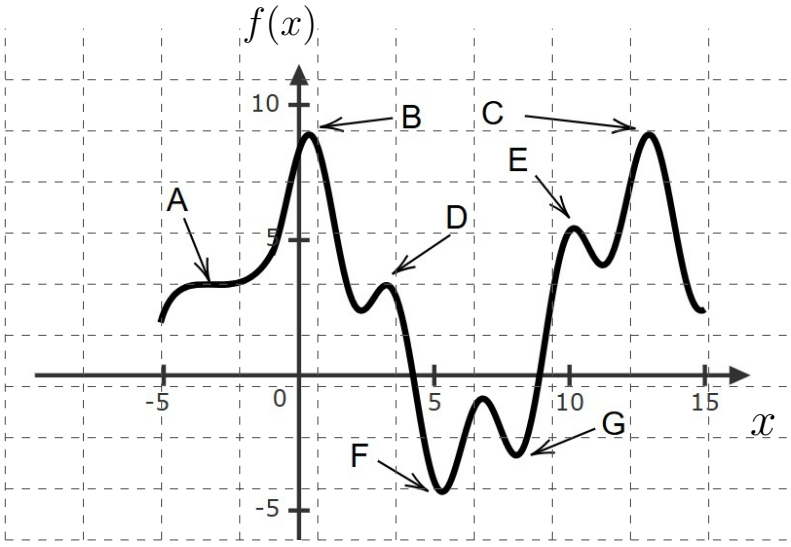


Figure Q1.1: Critical points of a function $f(x)$.

(b) Consider the following constrained optimization problem:

$$\begin{aligned} \underset{\boldsymbol{x}}{\text{minimize}} \quad & f(\boldsymbol{x}) = 5|x_1| + 2|x_2|, \\ \text{subject to} \quad & \boldsymbol{A}\boldsymbol{x} \leq \boldsymbol{b}, \\ & -x_1 \leq 0, \\ & -x_2 \leq 0, \end{aligned}$$

where $\boldsymbol{A} = \begin{bmatrix} 10 & 4 \\ 1 & 1 \end{bmatrix}$, $\boldsymbol{b} = \begin{bmatrix} 20 \\ 10 \end{bmatrix}$.

i. Draw a sketch of the feasible region and level curves of $f(\boldsymbol{x})$ in the same diagram. **[5 marks]**

ii. Geographically find the solution of the optimization problem given in question Q1.(b). **[4 marks]**

iii. Propose a matrix $\boldsymbol{A}$ and a vector $\boldsymbol{b}$ such that the optimization problem given in question Q1.(b) has a solution with either $x_1 = 0$ or $x_2 = 0$. **[6 marks]**

(c) Consider the function $f(\boldsymbol{x}) = 2x_1^2 - 6x_1 + \dfrac{x_2^3}{3} - \dfrac{9x_2^2}{2} + 20x_2 + 9$. Identify critical points and determine whether they are minima or maxima or neither minimum nor maximum. **[5 marks]**

Continued ...

Question 2.

(a) Consider a constrained optimization problem in which the variable x is bounded by $-5 \leq x \leq 8$ where $x \in \mathbb{R}$. Which of the transformation(s) shown in Figure Q2.1, can be used to convert the constrained problem into an unconstrained one? Justify your choice. **[6 marks]**

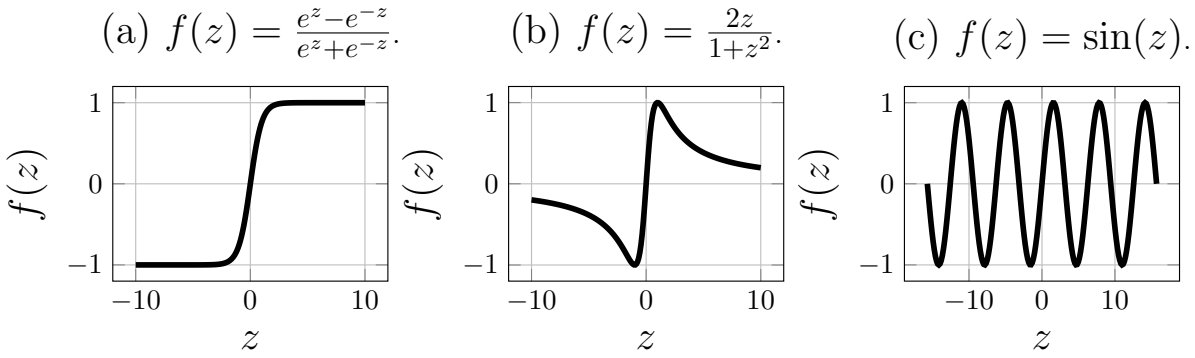


Figure Q2.1: Functions for transformation.

(b) Consider the constrained optimization problem:

$$\begin{aligned} \underset{\boldsymbol{x}\in\mathbb{R}^n}{\text{minimize}} \quad & f(\boldsymbol{x}) = \frac{1}{2}\|\boldsymbol{x}\|_2^2, \\ \text{subject to} \quad & y_i\big(\boldsymbol{a}_i^\top \boldsymbol{x} + b_i\big) = 1, \quad i = 1, \ldots, m, \end{aligned}$$

where $y_i \in \{-1, +1\}$ and $\boldsymbol{a}_i \in \mathbb{R}^n$ are given. Here, $\boldsymbol{x} \in \mathbb{R}^n$ and $b_i \in \mathbb{R}$ are optimization variables.

i. Show that the optimal solution is equivalent to

$$\boldsymbol{x}^* = \sum_{i=1}^{m} \lambda_i \, y_i \, \boldsymbol{a}_i.$$

Here, λ_i is the Lagrange multiplier. **[6 marks]**

ii. Show that $\displaystyle\sum_{i=1}^{m} \lambda_i \, y_i = 0$. Here, λ_i is the Lagrange multiplier. **[3 marks]**

iii. Let $\boldsymbol{\lambda} = [\lambda_1, \ldots, \lambda_m]^\top$, $\boldsymbol{y} = [y_1, \ldots, y_m]^\top \in \mathbb{R}^m$, $\boldsymbol{A} = [\boldsymbol{a}_1, \ldots, \boldsymbol{a}_m] \in \mathbb{R}^{n\times m}$. Show that the value of the objective function at the optimum can be expressed as

$$f(\boldsymbol{x}^*) = \frac{1}{2}\|\boldsymbol{x}^*\|_2^2 = \frac{1}{2}(\boldsymbol{\lambda} \odot \boldsymbol{y})^\top (\boldsymbol{A}^\top \boldsymbol{A})(\boldsymbol{\lambda} \odot \boldsymbol{y}),$$

where $\odot$ denotes the element-wise (Hadamard) product. **[10 marks]**

Continued ...

Question 3.

A cloud service provider offers computing resources to three categories of users, namely as Type A, Type B, and Type C, based on their computational needs. The monthly resource demands of each user type are as follows: Type A users require 8 GB of RAM, one GPU, and 100 GB of storage. Type B users require half of Type A's RAM requirement, half of Type A's storage requirement and three GPUs. Type C users require twice of Type A's RAM requirement, half of Type B's storage requirement and one GPU.

Suppose that number of users for types A, B and C are x_A, x_B and x_C, respectively. The cloud service provider aims to maximize its monthly revenue, with subscription fees for Type A, B, and C users set at Rs. 20,000, Rs. 35,000, and Rs. 25,000 per month, respectively. The provider has a limited pool of computing resources to allocate on a monthly basis. It has a total of 200 GB of RAM, 50 GPUs, and 1500 GB of storage. These resources must be shared among the three types of users. It is required that all available RAM and GPUs be fully utilized.

(a) Formulate an optimization problem to find the number of users of types A, B and C that maximize the monthly revenue of the service provider. Note that it is required to provide service to all three user types. **[5 marks]**

(b) Write down the Lagrangian function and KKT conditions (primal feasibility, dual feasibility and complementary slackness). **[8 marks]**

(c) Use the KKT conditions to solve the optimization problem. Find the feasible values of x_A, x_B and x_C, which maximize revenue (In your calculations, for the sake of simplicity, ignore the conditions that values of x_A, x_B, and x_C are non-negative). **[12 marks]**

Continued ...

Question 4.

(a) Consider the constrained optimization problem:

$$\begin{aligned} \underset{\boldsymbol{x}}{\text{minimize}} \quad & ax_1 + bx_2, \\ \text{subject to} \quad & x_1 \geq c, \\ & x_2 \geq d. \end{aligned}$$

Suppose that this problem is transformed to a unconstrained problem using quadratic penalty formulation with penalty parameter ρ (> 0) which is updated at $k+1$-th iteration as $\rho^{k+1} = \mu\rho^k$, starting from $\rho^0 = 5$.

i. Assuming the initial point lies in the infeasible region, and $a, b > 0$, show that the solution of the unconstrained problem obtained using the quadratic penalty method is

$$x_1 = c - \frac{a}{2\rho} \text{ and } x_2 = d - \frac{b}{2\rho}.$$

[6 marks]

ii. Let the solution of the transformed problem at iteration k be x_k, and the optimal solution of the constrained optimization problem given in question Q4.(a) be x^*. For $\mu = 1.05,\ a = 5,\ b = 10,\ c = 4$ and $d = 6$. How many iterations are needed to satisfy $\|x_k - x^*\|_2^2 \leq 10^{-3}$. **[6 marks]**

(b) Consider the constrained optimization problem given below:

$$\begin{aligned} \underset{\boldsymbol{x}}{\text{minimize}} \quad & 5x_1^2 + 4x_1x_2 + x_2^2, \\ \text{such that} \quad & 5x_1^2 + 2x_2^2 = 14, \\ & 3x_1 + 2x_2 \geq 5. \end{aligned}$$

i. What are the primal and dual optimization problems? **[4 marks]**

ii. What is the dual function? **[2 marks]**

(c) The first and second Wolfe conditions are given by

$$f(\boldsymbol{x}^{(k+1)}) \leq f(\boldsymbol{x}^{(k)}) + \beta\alpha\nabla_{\boldsymbol{d}^{(k)}} f(\boldsymbol{x}^{(k)}) \text{ with } \beta \in [0,\ 1],$$

$$\nabla_{\boldsymbol{d}^{(k)}} f(\boldsymbol{x}^{(k+1)}) \geq \sigma\ \nabla_{\boldsymbol{d}^{(k)}} f(\boldsymbol{x}^{(k)}) \text{ with } \sigma \in [0,\ 1].$$

It is given that $f(\boldsymbol{x}) = x_1 + x_1x_2 + x_1^2$, $\boldsymbol{x}^{(k)} = [1,\ 2]^T$, $\boldsymbol{d}^{(k)} = [-1\ -2]$, $\beta = 1\times10^{-4}$ and $\sigma = 0.9$.

i. What are the primary purposes of the first and the second Wolfe conditions? **[2 marks]**

ii. Is step size $\alpha = 2.5$ feasible? Justify your answer. If it is not feasible, what is your next step? **[5 marks]**

- End of Question Paper -